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- *Composite Propellant Ignition Fundamentals*
- *Pyrogen Compositions and Safety*
- *Igniter Testing Process and Test Results*
- *Igniter Sizing for Composite Motor Start Up*

Composite Propellant Ignition Fundamentals

The best pyrogens for solid motors have the following characteristics:

- High radiant heat flux per gram to bring the propellant surface up to self-ignition temperature.
- High gas generation to pressurize the rocket motor core above the motor's critical pressure for sustained combustion.
- Fast action time from electrical firing of the igniter to the point of maximum heat flux and pressure.
- Low sensitivity to ambient pressure and temperature.
- Minimal shock to the surface of the propellant to reduce risk of fracture.
- Low sensitivity to static discharge of ingredients and final igniter form.

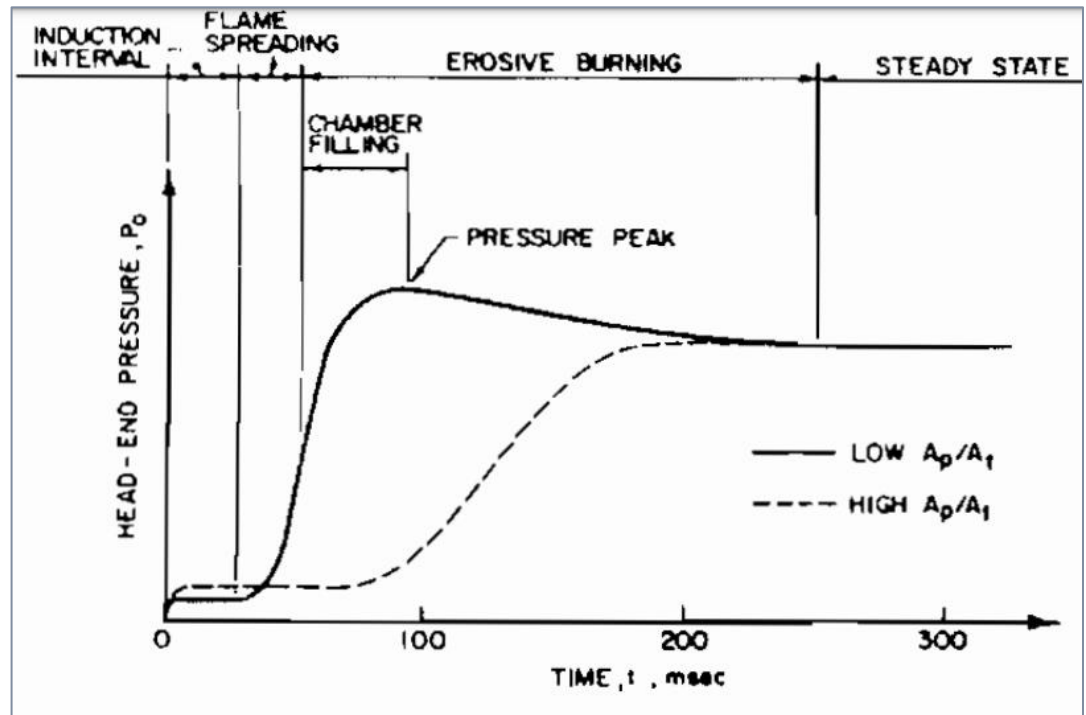
Especially important for upper stage ignition:

- Reliable, fast ignition at low ambient temperature and pressure.

Transient Phases of Solid Rocket Motor Startup

After the point when the electrical circuit is closed...

1. Electrical delay
2. Induction
3. Flame Spreading
4. Chamber Filling
5. Erosive Burning
6. Steady State



Ignition is enhanced by the use of a Burst Disk at the nozzle:

- Faster rise time to critical pressure and steady state combustion.

Important Ignition Parameters

Critical igniter sizing and motor design effects...

- Amount of heat transferred to the grain surface per unit of time (Heat Flux)
 - Typically >40 cal/cm² per second, higher for instant-on
- Minimal pressure for sustain steady-state combustion (Critical pressure)
 - Typically 50-100 psi for high-performance propellants
 - For low-performance propellants or instant-on: >150 psi
- Secondary effects of core resonance:
 - Long motors
 - Low port/throat area ratio
 - Erosive burning
 - Vehicle flight resonance or high acceleration

NOTE...

Depends on: propellant ignitability, grain design, and initial Kn/pressure

Igniter Pyrogen Safety Requirements

When choosing a pyrogen formula and making the igniter...

- Readily accessible ingredients with minimal safety hazards
- Proper safety procedures for combining ingredients and making the igniter
- Chemically stable in final form over storage time
- Minimal susceptibility to static discharge, friction, and impact
- Made easily onsite if regulatory issues become a concern

Understand the risks, including, but not limited to, the following items:

- Use appropriate personal protection equipment
- Use proper handling procedures
- Minimize the batch size
- Keep incompatible chemicals and equipment away from work area

Electrical Ignition Options

*A quick overview of physical electrical starters...
...(not within the scope of this presentation)*

- Low energy vs. High energy
- Wrapped nichrome head
- Bridge wire
- E-Match
- Conductive pre-dip

Goals:

- *Electrically safe when installed in the motor*
- *Compatible with launch system's continuity check*

Typical types of Pyrogen Compositions

Basics: Oxidizer, Fuel, Binder

Possible Additives: Catalyst, desensitizers, crazy molten stuff

General classes of pyrogens...

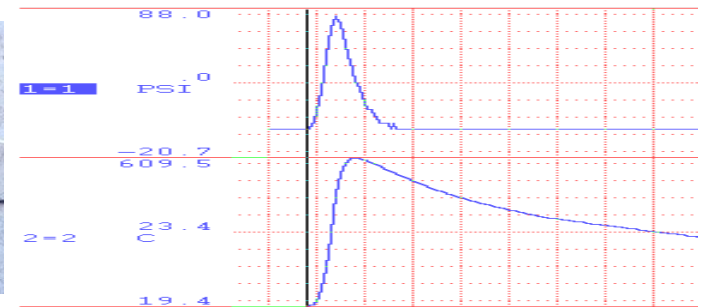
- BP derivatives
- First Fires (classic pyrotechnics initiators)
- Thermites (an elemental metal plus a transition metal oxide)
- High-performance mil-spec compositions
- “I tried this combination of stuff and it seems to work!”

→ *How do we objectively compare the performance and trade-offs?*

→ *How do we choose a composition which meets our requirements?*

Igniter Testing Apparatus

- Short 75mm snap-ring casing as a test chamber
 - 6061-T6 aluminum tube 5.75" long and with a 2.75" internal diameter
- Tapped forward closure with a pressure transducer
 - 1000 PSI Measurement Specialist M3431-000006-01KPG
- Aft closure with a means of inserting a sealed igniter-under-test
 - 1/4" NPT pressure fitting with igniter leads potted with epoxy
- High-temperature thermocouple in closer proximity to the igniter
 - K-type, 1000°C maximum range
- Data acquisition system with sufficient sample rate and resolution
 - Dataq data acquisition module and a Tektronix TDS2004C oscilloscope



Pyrogen Parameters from Test Data

Directly Measured Parameters

- Peak pressure
- Time to peak pressure
- Thermocouple temperature

Derived Parameters

- Characteristic Impetus (work/gram of gas generation)
- Estimate of relative Heat Flux, in calories per cm² per second
- Hot mass flow, estimated from the average temperature and mass/time
- Relative Heat Flow Density:
The pyrogen's ability to deliver both heat flux and hot mass flow

Explanation of the Derived Measures

Characteristic Impetus

The Impetus value (W) may be used to calculate the amount of igniter pyrogen (M) needed for a desired peak pressure (P) and for the core volume (V) of a solid rocket motor. It's the work per gram in generating gas volume.

$$W = P \ V / M \quad \text{(psi-in}^3\text{/gram)}$$

Heat Flux

A rough estimate of relative Heat Flux (Φ), in calories per cm² per second, was calculated by:

$$\Phi = k_E \ W \ M / t_A / (2 \pi \ R \ L) \quad \text{(cal/cm}^2\text{-sec)}$$
$$k_E = 0.238 \text{ (cal/J)}$$

Effective igniter length (L in cm), thermocouple distance (R in cm) the Impetus (W in J/gram), the pyrogen mass (M in grams), and the action time (t_A , time to peak pressure in seconds). Heat energy per unit area per unit of time.

Igniter Preparation

The same type of electrical ignition heads were used for all pyrogen tests:

- 5 turns of 30-AWG nichrome over 1" of one conductor of 18/2 solid-core wire
- About four wraps of Teflon plumbers tape was applied over this 1" end before wrapping the nichrome; this prevents shorting through the PVC insulation.
- A thin coating of nitrocellulose lacquer was applied over the completed igniter head as a sealant and initiator.
- The final diameter of the igniter head was approximately 0.07" to 0.08" before applying the pyrogen. The nominal resistance of the heads was 1.2 ohms.



The pyrogen was applied to the heads by either dipping or molding:

- The mass of the pyrogen was recorded for each igniter.
- For the dry samples, they were contained in thin plastic film and lit with an electric match centered in the pyrogen powder.

Test Procedure

The following procedure was performed for each igniter pyrogen type:

- The rear closure was cleaned and the igniter sample with thermocouple was inserted and re-assembled. The forward closure with pressure transducers was cleaned and re-assembled.
- The electrical connections and transducer were powered on and verified for ambient levels. Then, the data acquisition system was started, followed by the firing system being armed and fired.
- The data from both acquisition sources were saved to disk on the host laptop computer.
- The firing system was disarmed and the electrical power was removed.

A minimum of three tests were done for each type of pyrogen.

Test Results

The results of the series of pyrogen tests are shown in the following table:

Pyrogen	Peak Pres (psi)	Time to Peak Pres (msec)	TC Temp (°C)	Impetus (psi-in ³ /gm)	Heat Flux (cal/cm ² /sec)	Temp × Mass Flow (°C-gm/sec)	Normalized Heat Flow Density
BKNO3-Viton	105	67	840	5000	69.4	2724	1.00
BKNO3-NC	102	73	768	4800	66	2200	0.77
Mg-Teflon-Viton	103	50	930	4700	110	1488	0.87
Mg-Teflon(dry)	106	40	>1000*	3000	109	3429*	1.98*
Mg-KNO3-epoxy	53	133	476	2400	20.3	626	0.067
Al-CuO thermite	45	184	277	2038	12.1	183	0.012
First fire	105	120	594	2200	29.2	674	0.104
BP-resin pellet	56	130	242	2570	44.6	693	0.164

** Note: The dry Mg-Teflon results were difficult to measure due to the fast action time and the off-scale thermocouple measurements.*

Discussion of Results

1. The Boron-based igniters performed the highest in both heat and gas generation.
2. The MTV igniters performed nearly as well but have some practical limitations.
3. The Mg + Teflon without a binder is a flash powder which likely would damage the grain.
4. The thermite powder did not perform well as a self-contained pyrogen. As expected, thermite produces mostly hot solids and liquids with less radiant heat and lowest gaseous output. Thermite relies on blasting the propellant surface to generate pressurization gas. Risks: grain damage, and difficulty sizing.
5. The remaining pyrogen types, which included typical “motor starters” used in hobby rocketry, placed significantly lower in performance than the upper three in all the important measures:

Gas Generation, Heat Flux, and the Action Time.

Notes on Electrostatic Sensitivity

Pryogen	Mix Ratios (by mass)	Static Sensitivity Rank (joules) (1 = low sensitivity)
B – KNO ₃ – V (dip/mold)	25 : 60 : 15	1 (>5)
B – KNO ₃ – NC (pressed)	24 : 71 : 5	5 (0.124)
Mg – Teflon - Viton	54 : 30 : 16	6 (0.03)
Mg – Teflon	50 : 50 (dry)	8 (0.005)
Mg – KNO ₃ – epoxy	15 : 60 : 25	2 (1.25)
Al – CuO thermite	4.4 : 1 (dry)	7 (0.020)
First Fire	(various)	3 (0.875)
BP-resin pellet	(Pyrodex, etc)	4 (0.225)

Note: Additional data were gathered from online sources for NASA, military, and commercial motor ignition systems.

Boron – Potassium Nitrate - Viton (BPNV) Igniter Composition

Ratio by mass: B=25 : KNO₃=60 : Viton=15

Boron – amorphous (non-crystallized), brown fine powder (<2um), >=92% pure

Potassium Nitrate (KNO₃) – white fine-milled powder, finer the better

Viton – raw (uncured) fluoroelastomer (FKM) polymer, ~66% fluorine content

How I mix the pyrogen:

1. Dissolve Viton in acetone (begin with 10:1 Acetone:Viton)
2. Add boron powder and mix thoroughly
3. Add potassium nitrate slowly, and mix thoroughly
4. Use more acetone to get the desired consistency...
 - For dipping, a thin paste
 - For pellet casting, a thick paste
5. Can be reconstituted with acetone later if dry

Igniter Pyrogen Sizing

INPUTS:

D = the diameter of motor's core at the top grain in inches.

L = the length of the motor's propellant core in inches.

P = the desired pressure developed in the core in PSI (typically use 200-300 PSI).

W = the "impetus" (ability to generate gas) of the pyrogen in PSI-in³/gram.

For example, use 5000 for BKNO₃-V. See test results for others.

OUTPUTS:

M = mass of the igniter pyrogen in grams.

S = size of the igniter pyrogen's surface length in inches (for dipped igniters).

N = number of pellets (for pellet basket igniters).

CALCULATIONS:

Find the core volume: $V = \pi * D^2/4 * L$ (in³)

Find the pyrogen mass: $M = V * P / W$ (grams)

Find the pyrogen size for dipped igniters and the number of pellets for basket igniters:

$S = D * 1.5$ (pyrogen length in inches for dipped igniters, or length of pellet basket)

$N = M / q$ (q is the mass per pellet: q=0.42 grams for 1/4" pellets, for example)

BPNV Igniter Pyrogen Sizing Examples

EXAMPLE 1: A “J” motor, core diameter of 0.5” and a core length of 12”. Ignition at 300 psi.

$$V = 3.14159 * (0.5)^2/4 * 12 = 2.36 \text{ in}^3$$

$$M = 2.36 * 300 / 5000 = 0.14 \text{ grams}$$

→ $S = 0.5 * 1.5 = 0.75$ ” (length of pyrogen for dipped igniter).

EXAMPLE 2: An “M” motor, core 0.75” dia, core length of 40”. Ignition pressure of 200 psi.

$$V = 3.14159 * (0.75)^2/4 * 40 = 17.7 \text{ in}^3$$

$$M = 17.7 * 200 / 5000 = 0.71 \text{ grams}$$

→ $S = 0.75 * 1.5 = 1.13$ ” (length of pyrogen for dipped igniter).

EXAMPLE 3: An “O” motor, core dia of 2”, a core length of 33”. Ignition pressure of 400 psi.

$$V = 3.14159 * (2)^2/4 * 33 = 103.7 \text{ in}^3$$

$$M = 103.7 * 400 / 5000 = 8.3 \text{ grams}$$

→ $S = 2 * 1.5 = 3.0$ inches (length of dipped pyrogen or pellet basket).

→ $N = 8.3 / 0.42 = \sim 20$ pellets (1/4” dia by 1/4” tall).

Conclusions

1. The closed chamber tests of various igniter pyrogens show that dipped or molded BKNO₃ using Viton as a binder is at least as good as pressed BKNO₃ with a resin or nitrocellulose binder.
2. BKNO₃ and MTV are in a class of their own above the other pyrogens tested.
3. The Mg-Teflon™ pyrogens (dry or with Viton as a binder) were both high performance, but should be avoided due to the risks of handling the static-sensitive dry ingredients.
4. Thermites do not act as propellant igniters within the essential requirements (detailed earlier). Thermite causes the partial decimation of the propellant surface to produce pressurization gas.
5. The results of these experiments provide practical evidence to justify using Boron + Potassium Nitrate + Viton (BPNV) as a dipped or molded pyrogen for fast, reliable solid motor ignition.

... Questions ???



Video link:

<https://photos.app.goo.gl/x22nEssS3PW1r7hc9>



1/2 gram of dipped BPNV.
~60 msec action time.
See igniter clips for scale.